ARJAN SINGH PUNIANI

NEURAL ENGINEERING · BRAIN-COMPUTER INTERFACES · TRANSLATIONAL NEUROTECHNOLOGY

Vallejo / Napa, CA · (559) 392-1572 · arjan.singh.puniani.main@gmail.com · arjansinghpuniani.com

M.S. BIOENGINEERING	PROVISIONAL PATENT	PUBLICATIONS	FIRST AUTHOR	NSF I-CORPS
GPA 4.0	**SeizeFreeze filed**	**2 peer-reviewed papers**	**BCI manuscript**	**Fellow**

01 / HUMAN NEURAL ENGINEERING

Rehab Neural Engineering Labs (RNEL), University of Pittsburgh — Pittsburgh, PA · 2022-2024

R&D Neural Engineer / BCI Researcher | **HUMAN PARTICIPANTS · INTRACORTICAL MICROSTIMULATION · 3,024 DOCUMENTED HOURS**

- Built and studied psychophysical calibration paradigms for intracortical microstimulation of somatosensory cortex in implanted human BCI participants; developed a Pygame Space-Invaders-like two-alternative forced-choice task while preserving stimulus timing, experimental control, and tactile detection-threshold estimation.
- Participants rated the gamified task significantly higher for aesthetics, motivation, and reward, while average detection thresholds did not differ significantly from conventional calibration; findings were submitted to the Journal of Neural Engineering.
- Developed real-time neural interfaces integrating electrophysiology, virtual environments, and assistive robotic control; analyzed 940+ longitudinal calibration sessions and identified systematic signal degradation that prompted broader investigation.

Sensorimotor Integration Laboratory, University of Pittsburgh — Pittsburgh, PA · 2021-2022

Neurophysiology Data Analyst & Machine Learning Engineer

- Developed a MATLAB microsaccade detector for nonhuman-primate eye tracking and a Hidden Markov Model for laminar-probe neural activity, identifying task-related microbursts with 81% accuracy.

02 / TRANSLATIONAL NEUROTECHNOLOGY

SeizeFreeze / Cyberpunk Reality, Inc. — California · 2022-Present

Founder + Neurotechnology Device Lead · NSF I-Corps Fellow | **PROVISIONAL PATENT FILED · $12.5K DOCUMENTED AWARDS**

- Invented and filed a provisional patent for SeizeFreeze, a proposed closed-loop focal cortical-cooling neuro-implant for drug-resistant focal epilepsy; developed thermal models, thermoelectric device architecture, materials and packaging constraints, sensing/control concepts, and neurosurgical integration requirements.
- Selected as an NSF I-Corps Fellow and funded to conduct market validation; interviewed patients with epilepsy, neurosurgeons, FDA regulators, and hospital procurement stakeholders to test clinical need, workflow, regulatory, and commercialization assumptions. Advanced the concept through prototype planning and documented innovation awards while keeping therapeutic claims bounded to available evidence.

03 / COMPUTATIONAL + TRANSLATIONAL SYSTEMS

ReasonOS / VECTOR ECG / Independent research + working software · 2026

Built a replayable, provenance-aware architecture for representing expert reasoning as explicit state changes - observation, measurement, competing hypotheses, contradiction, revision, and review - with deterministic traces and visible uncertainty boundaries.

Motorsport Neurotrauma / ICMS / independent project · 2026

Developed an exploratory crash-to-neurotrauma documentation framework linking mechanism, occupant-protection context, acute neurologic red flags, and initial disposition; revised it after critique from track-experienced motorsport physicians.

Vector Tennis / Interactive physics playground · 2026

Built an arcade-scale tennis simulation in which timing, racket position, pace, spin, Magnus-like curve, and spin-sensitive bounce remain causally linked; explicitly labels telemetry as game-relative rather than sports-science measurement.

04 / CLINICAL RESEARCH

UCSF Eureka Platform — San Francisco, CA · 2021

Clinical Research Coordinator | **960 DOCUMENTED HOURS**

- Translated investigator study plans and informed-consent materials into participant-facing digital trial workflows; partnered with engineers and product managers while maintaining HIPAA requirements and study-data quality.
- Authored training and onboarding materials for study teams and developed participant-focused research content and operational documentation for digital clinical-trial deployment.

05 / EARLIER OPERATING + FINANCIAL EXPERIENCE

Rigetti Computing / Chief of Staff · Berkeley, CA · 2013-2015

Early-stage operating role at a quantum-computing company; supported Y Combinator fundraising, investor materials, and patent/IP efforts.

Goldman Sachs / Private Wealth Management Summer Intern · San Francisco, CA · 2012

Built a dividend discount model to support investment analysis for ultra-high-net-worth individuals within Private Wealth Management.

06 / EDUCATION + SCHOLARSHIP

University of Pittsburgh, Swanson School of Engineering · M.S., Bioengineering - Brain-Computer Interfaces · GPA 4.0 · 2023

University of California, Berkeley · B.S., Chemical Engineering · 2013

PEER-REVIEWED Wiest MC, Puniani AS. "Conscious active inference I: A quantum model naturally implements the path integral needed for real-time planning and control." Comput Struct Biotechnol J. 2025;30:108-121.

PEER-REVIEWED Wiest MC, Puniani AS. "Conscious active inference II: Quantum orchestrated objective reduction among intraneuronal microtubules naturally accounts for discrete perceptual cycles." Comput Struct Biotechnol J. 2025;30:94-107.

FIRST AUTHOR Puniani AS, Verbaarschot C, Greenspon CM, Higgins H, Bensmaia SJ, Gaunt RA. "Making psychophysical calibration tasks for brain-computer interfaces more fun..." *Manuscript submitted to Journal of Neural Engineering.*

07 / METHODS + SCIENCE COMMUNICATION

METHODS ICMS · psychophysics · electrophysiology · human-subject research · Python · MATLAB · C++ · R · HMMs · ANSYS · SolidWorks · Unity · Git

SCIENCE COMMUNICATION **Physics World contributor ·** "Novel decoder helps people with paralysis click-and-drag a computer cursor using just their thoughts" (2021) · "Deep neural networks track eye movements during MRI scans" (2022)